

⊗ BFS → level-wise traversal
 ⊗ DFS → depth-wise traversal
 } Blind search/uninformed search

⊗ Dijkstra → SS SP
 Bellman-Ford
 } path-finding algorithms

heuristic = no. of misplaced tiles
 = manhattan distance
 = euclidean
 Goal state

BFS → b^d
 DFS → b^m

exponential time complexity

Initial State

1	2	3
5	-	8
4	6	7

$h = 5$

1	2	3
4	5	6
7	8	-

$h = 0$

up

down

left

right

1	-	3
5	2	8
4	6	7

$h = 6$

1	2	3
5	6	8
4	-	7

$h = 5$

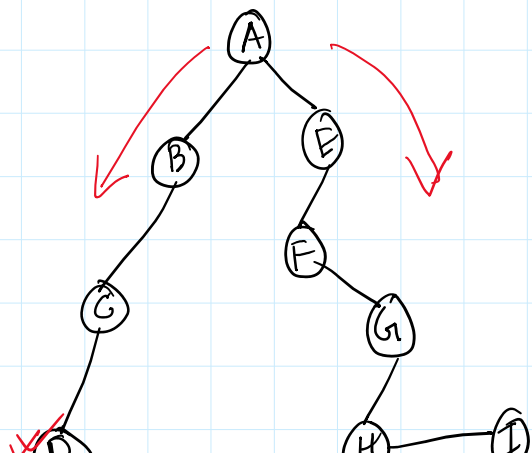
1	2	3
-	5	8
4	6	7

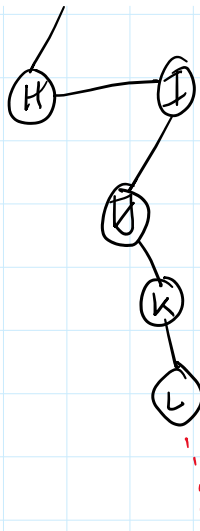
$h = 4$

1	2	3
5	8	-
4	6	7

$h = 5$

Blind search/uninformed search → no information where goal is





How can we make it efficient?

→ More information regarding approximated distance from goal state.

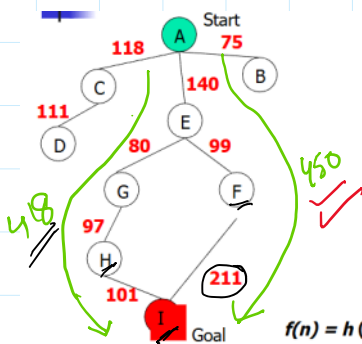
↪ heuristic (h)

Informed Search

GBFS

A* Search

GBFS



$f(n) = h(n)$ = straight-line distance heuristic

State	Heuristic: $h(n)$
A	366
B	374
C	329
D	244
E	253
F	178
G	193
H	98
I	0

14

$$f(n) = h(n)$$

A³⁶⁶

A³⁶⁶ [C³²⁹ E²⁵³ B³⁷⁴]

E²⁵³ [C³²⁹ B³⁷⁴ G¹⁹³ F¹⁷⁸]

F¹⁷⁸ [C³²⁹ B³⁷⁴ G¹⁹³ I⁰]

path: $A \rightarrow E \rightarrow F \rightarrow I$
 $140 + 99 + 211 = 450$ (Sub-optimal path)

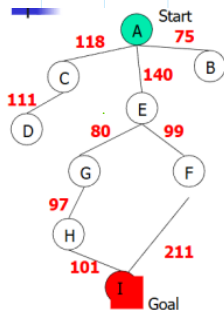
- * Not optimal
- * Not complete

$$F^{178} \begin{array}{|c|c|c|c|} \hline C^{329} & B^{314} & G^{193} & I^0 \\ \hline \end{array}$$

$$I^0 \begin{array}{|c|c|c|} \hline C^{329} & B^{314} & G^{193} \\ \hline \end{array}$$

A* Search

$f(n) = g(n) + h(n)$
 → cost from start to that node
 → approximated distance from that node to goal



State	Heuristic: $h(n)$
A	366
B	374
C	329
D	244
E	253
F	178
G	193
H	98
I	0

$f(n) = h(n)$ = straight-line distance heuristic

14

$$f(A) = g(A) + h(A) = 0 + 366 = 366$$

$$f(C) = 118 + 329 = 447$$

$$f(E) = 140 + 253$$

$$f(G) = 220 + 193$$

$$A^{366} \begin{array}{|c|c|c|c|} \hline C^{447} & E^{393} & B^{449} & \\ \hline \end{array}$$

$$E^{393} \begin{array}{|c|c|c|c|} \hline C^{447} & B^{449} & G^{413} & F^{417} \\ \hline \end{array}$$

$$G^{413} \begin{array}{|c|c|c|c|} \hline C^{447} & B^{449} & F^{417} & H^{415} \\ \hline \end{array}$$

$$H^{415} \begin{array}{|c|c|c|c|} \hline C^{447} & B^{449} & F^{417} & I^{418} \\ \hline \end{array}$$

$$F^{417} \begin{array}{|c|c|c|c|} \hline C^{447} & B^{449} & I^{418} & I^{450} \\ \hline \end{array}$$

$$I^{418} \begin{array}{|c|c|c|} \hline C^{447} & B^{449} & I^{450} \\ \hline \end{array}$$

$I \leftarrow H \leftarrow G \leftarrow E \leftarrow A$

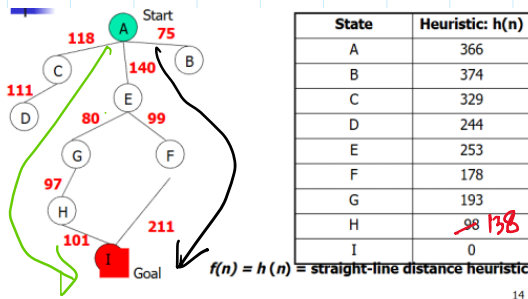
$$101 + 97 + 80 + 140$$

= 418 (Optimal path)

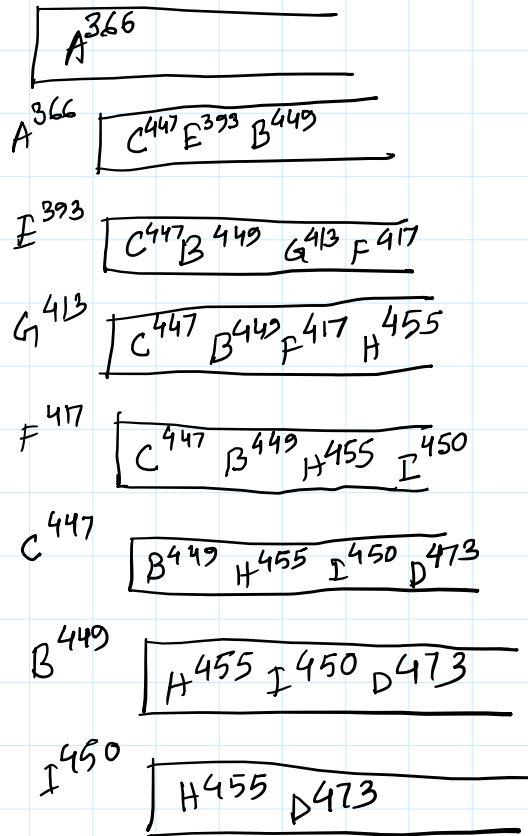
$A^* = \text{GBFS} + \text{UCS (Dijkstra)}$

* Admissibility: The estimated cost of going from a node to goal node must be lesser than actual cost.
 $h(n) \leq h^*(n)$, actual

to goal node must be lesser than actual cost -
 $h(n) \leq \underline{h^*(n)} \rightarrow \text{true/actual cost}$



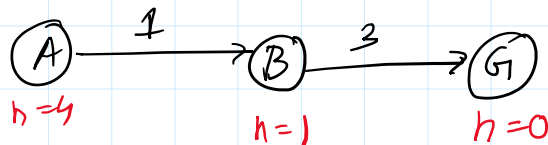
$I \leftarrow F \leftarrow E \leftarrow A$
 (Sub optimal path)



Optimality $\rightarrow$
 admissible heuristic

* Consistency: $\rightarrow$ triangular inequality

$$h(n) \leq c(n, n') + h(n')$$



n = current node
 n' = child node

$$h(A) \leq c(A, B) + h(B)$$

$$\Rightarrow 4 \leq 1 + 1 \rightarrow \text{inconsistent heuristic}$$

$$h(B) \leq c(B, G) + h(G)$$

Inconsistent
 but
 admissible

$$h(B) \leq C(B, G) + h(G)$$

$\Rightarrow \underline{f \leq 3 + 0} \rightarrow \text{consistent}$ "

but
 admissible
 heuristic

Admissible $\rightarrow$ Optimal solⁿ

Consistency $\rightarrow$ Efficient "

$\rightarrow$ All consistent heuristics are admissible

$\rightarrow$ However, not all admissible heuristics are consistent.

$$O(b^{GD})$$